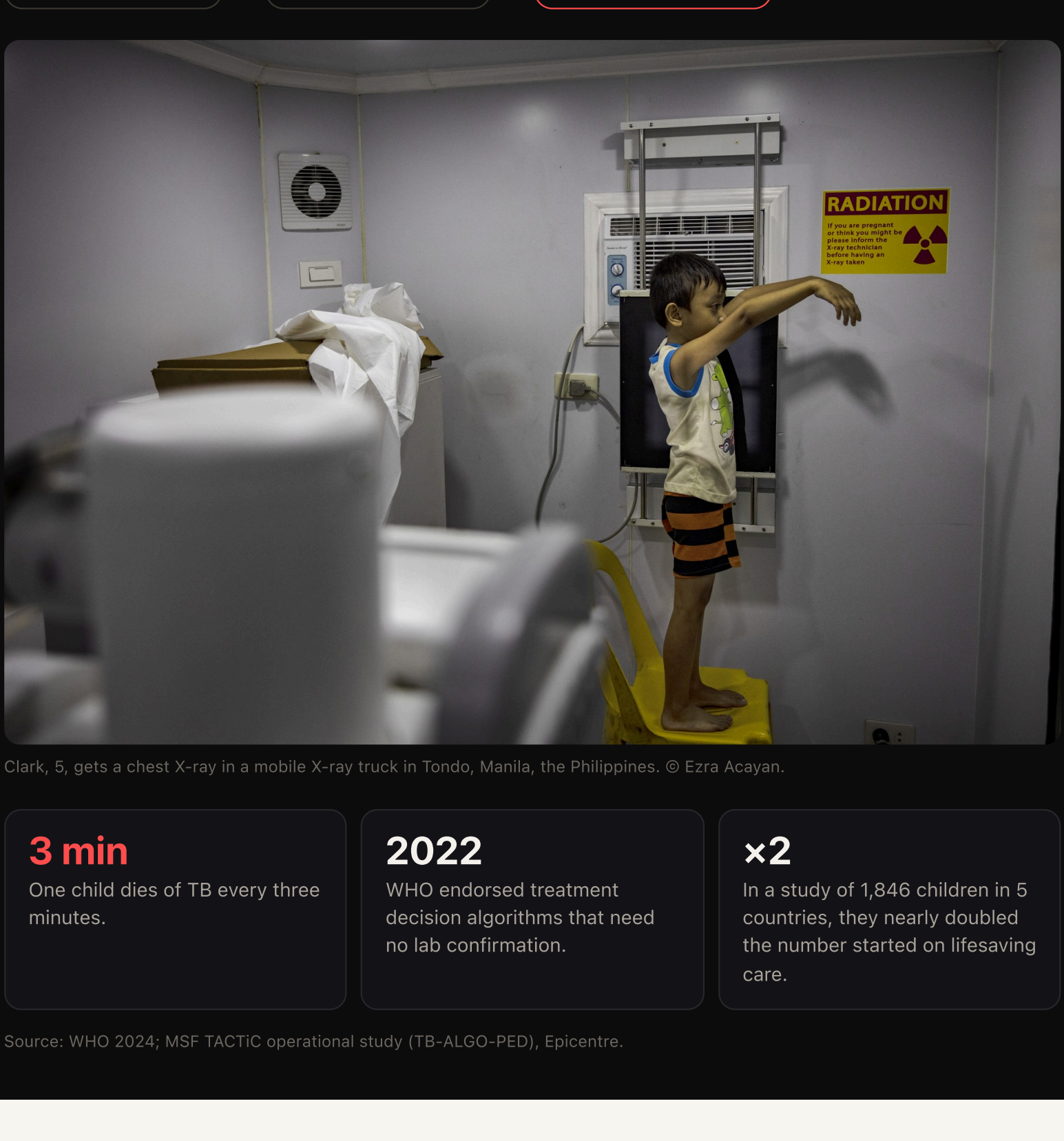
MSF TACTIC INITIATIVE · TB DIAGNOSIS IN CHILDREN

A child dies of TB *every 3 minutes.*

Almost none of them are diagnosed.

MORE DIAGNOSIS → MORE TREATMENT → MORE LIVES SAVED



Clark, 5, gets a chest X-ray in a mobile X-ray truck in Tondo, Manila, the Philippines. © Ezra Acayan.

3 min
One child dies of TB every three minutes.

2022
WHO endorsed treatment decision algorithms that need no lab confirmation.

×2
In a study of 1,846 children in 5 countries, they nearly doubled the number started on lifesaving care.

Source: WHO 2024; MSF TACTiC operational study (TB-ALGO-PED), Epicentre.

THE PROBLEM · 1 OF 2

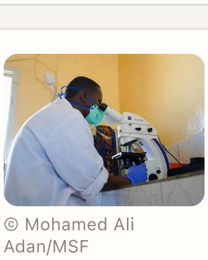
Biologically hard to detect

Four reasons the standard diagnostic toolkit, built around adults, fails children.

© Lynzy Billing/MSF

01 Non-specific symptoms

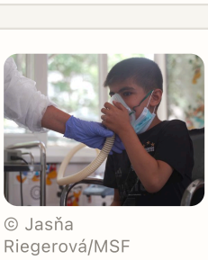
- Fever, fatigue, weight loss — TB symptoms in children **overlap with dozens of common illnesses**



© Mohamed Ali Adan/MSF

02 Lower bacterial loads

- Children have lower bacterial loads than adults
- Standard TB diagnostic tests designed for adults **frequently return false negatives**
- Children with TB go undiagnosed and untreated



© Jasňa Riegerová/MSF

03 Can't produce sputum

- Most diagnostics rely on sputum samples
- Most children under 5 years **can't cough up sputum on demand**

© Ezra Acayan/MSF

04 Extrapulmonary TB

- Children get TB in **lymph nodes, bones, brain and other non-lung sites** more frequently than adults
- Sample collection and bacteriological confirmation become harder

THE PROBLEM · 2 OF 2

Children are politically *neglected*

Public health systems and biotechnology prioritise what is scalable, profitable and measurable, resulting in a self-reinforcing cycle that keeps children with TB sidelined.

THE VICIOUS CYCLE

01 Standard TB lab tests miss children
Leading to misdiagnosis, delayed treatment and death.

↓

02 Children are undercounted in surveillance data
Which produces a lack of evidence.

↓

03 Less policy attention
So resources are not allocated to pediatric TB diagnosis.

↓

04 Less incentive to design better tests for children
Low diagnosis and continued invisibility feed straight back into step 01.

WHAT KEEPS IT TURNING

- Children are **less infectious than adults**, so pediatric TB is seen to have lower public health benefit
- Transmission-focused control strategies are driven by diagnosing and treating infectious adults
- Policy systems favour short-term, measurable outcomes
- Children can't vote or lobby, so only intermediaries can advocate for them

> **What breaks the cycle:** a diagnostic method that does not depend on adult-style laboratory confirmation. That method already exists and is endorsed by the WHO.

A PRACTICAL ALTERNATIVE

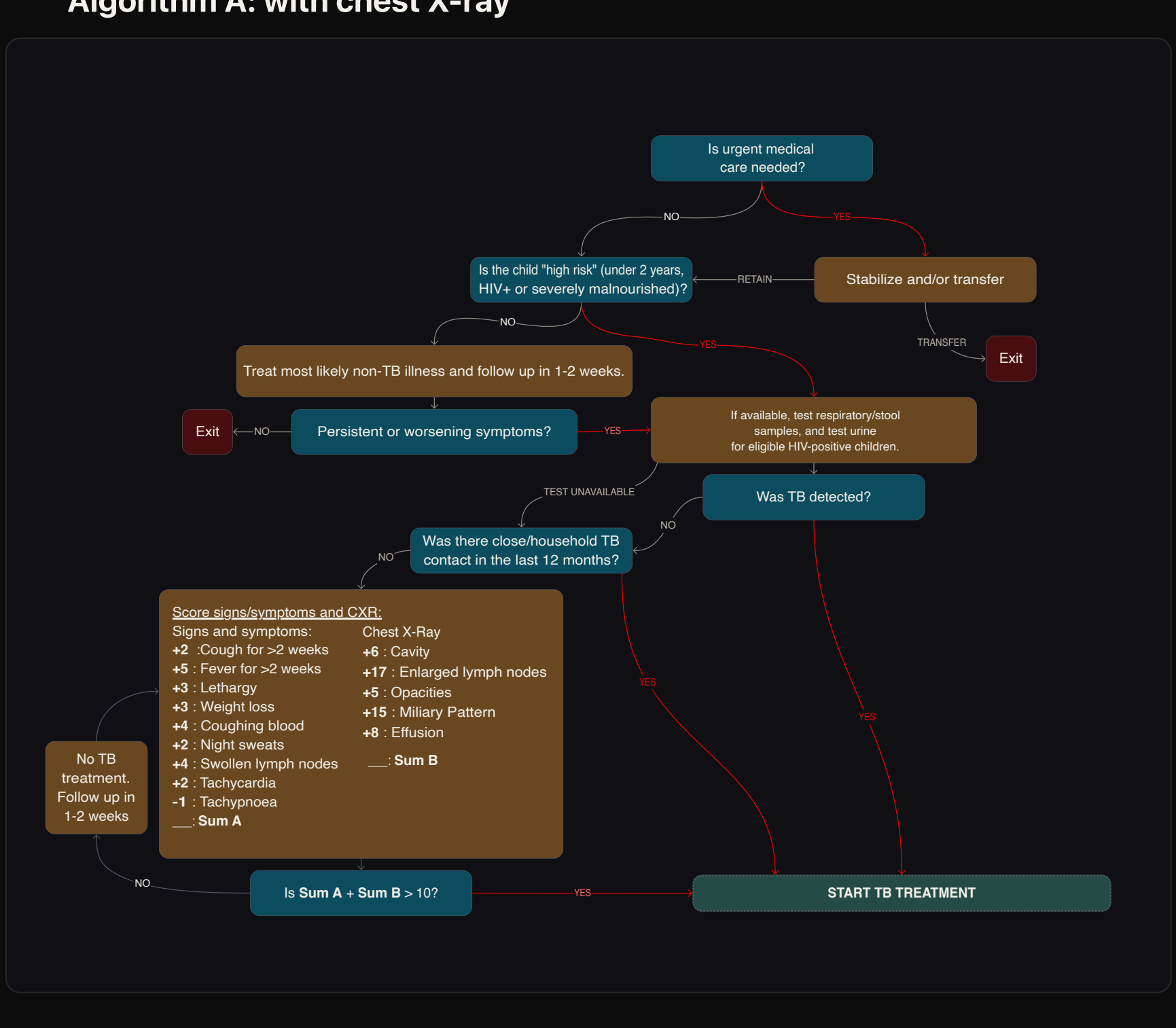
WHO Treatment Decision Algorithms: *a diagnostic flowchart that saves lives*

A scoring system that lets a clinician start TB treatment *without* laboratory confirmation.

- Identifies pulmonary TB in **children under 10 years** without advanced technology or specialists
- Systematises **non-specific symptoms**, TB exposure, basic test results and chest X-ray results (when available) into a cohesive scoring system
- Endorsed by the **World Health Organization** in 2022
- A score **above 10** triggers TB treatment — no lab confirmation required
- **Two versions:** **A** for clinics with chest X-ray, **B** for those without X-ray and/or radiological interpretation
- Based on data pooled from **thousands of children** across multiple countries

WHO · 2022

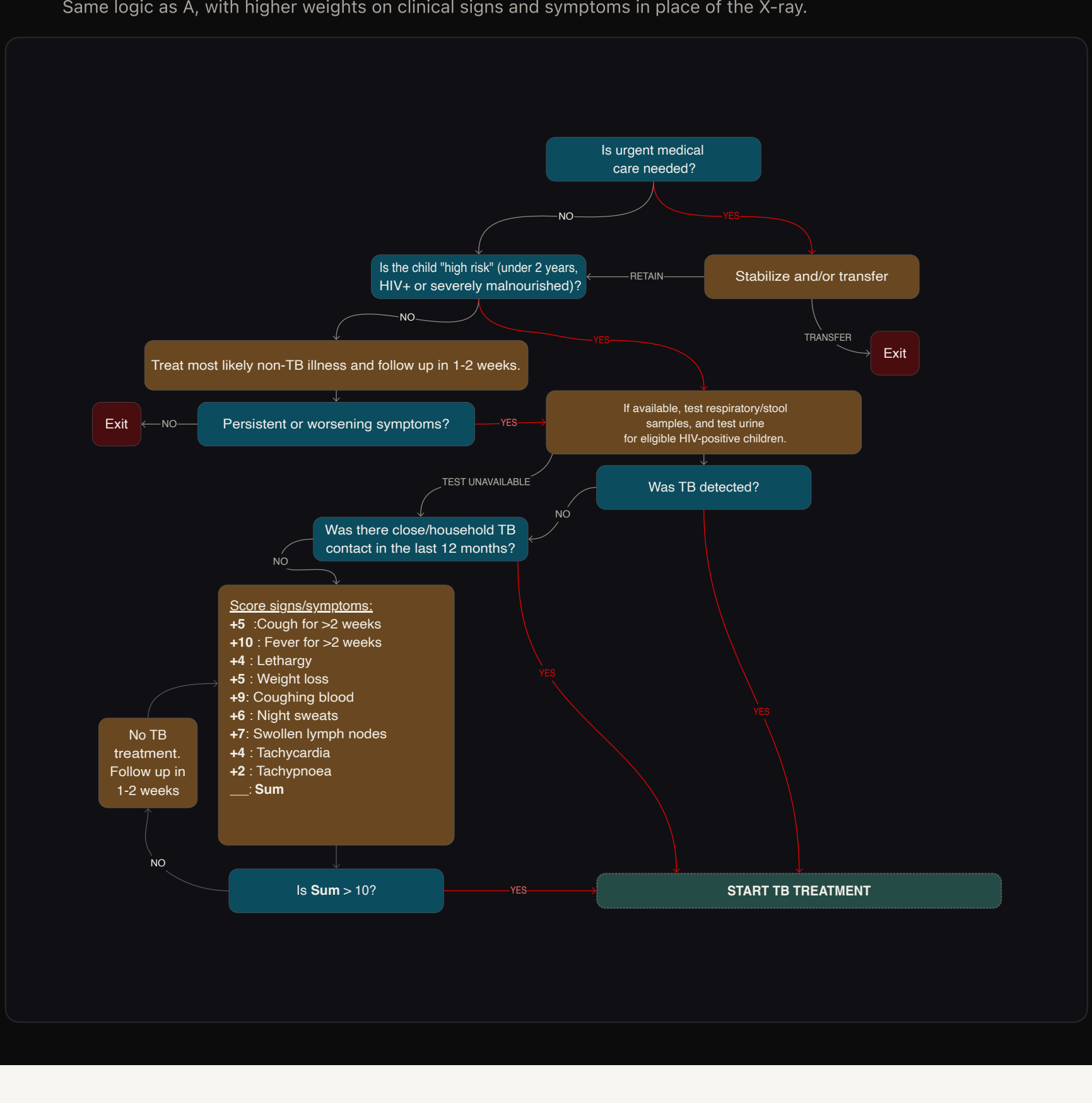
Algorithm A: with chest X-ray


Is urgent medical care needed?
NO
YES
Stabilize and/or transfer
RETAIN
TRANSFER
Exit
Is the child "high risk" (under 2 years, HIV+ or severely malnourished)?
NO
YES
Treat most likely non-TB illness and follow up in 1-2 weeks.
Persistent or worsening symptoms?
Exit
NO
YES
If available, test respiratory/stool samples, and test urine for eligible HIV-positive children.
TEST UNAVAILABLE
Was TB detected?
NO
YES
Was there close/household TB contact in the last 12 months?
NO
YES
Score signs/symptoms and CXR:
Signs and symptoms:
+2 :Cough for >2 weeks
+5 : Fever for >2 weeks
+3 : Lethargy
+3 : Weight loss
+4 : Coughing blood
+2 : Night sweats
+4 : Swollen lymph nodes
+2 : Tachycardia
-1 : Tachypnoea
___: Sum A
Chest X-Ray
+6 : Cavity
+17 : Enlarged lymph nodes
+5 : Opacities
+15 : Miliary Pattern
+8 : Effusion
___: Sum B
No TB treatment. Follow up in 1-2 weeks
NO
Is Sum A + Sum B > 10?
YES
START TB TREATMENT

WHO · 2022

Algorithm B: without chest X-ray

Same logic as A, with higher weights on clinical signs and symptoms in place of the X-ray.


Is urgent medical care needed?
NO
YES
Stabilize and/or transfer
RETAIN
TRANSFER
Exit
Is the child "high risk" (under 2 years, HIV+ or severely malnourished)?
NO
YES
Treat most likely non-TB illness and follow up in 1-2 weeks.
Persistent or worsening symptoms?
Exit
NO
YES
If available, test respiratory/stool samples, and test urine for eligible HIV-positive children.
TEST UNAVAILABLE
Was TB detected?
NO
YES
Was there close/household TB contact in the last 12 months?
NO
YES
Score signs/symptoms:
+5 :Cough for >2 weeks
+10 : Fever for >2 weeks
+4 : Lethargy
+5 : Weight loss
+9: Coughing blood
+6 : Night sweats
+7: Swollen lymph nodes
+4 : Tachycardia
+2 : Tachypnoea
___: Sum
No TB treatment. Follow up in 1-2 weeks
NO
Is Sum > 10?
YES
START TB TREATMENT

THE EVIDENCE

1,846 children. *5* countries.

MSF's TACTiC initiative ran the **TB ALGO PED** study to test the diagnostic accuracy and implementability of the WHO algorithms.

Guinea · Niger · Nigeria · South Sudan · Uganda

A child being evaluated for TB using the WHO algorithms in Uganda | © Stuart Tibaweswa | September 2024

THE THREE HIGH-RISK SUBGROUPS

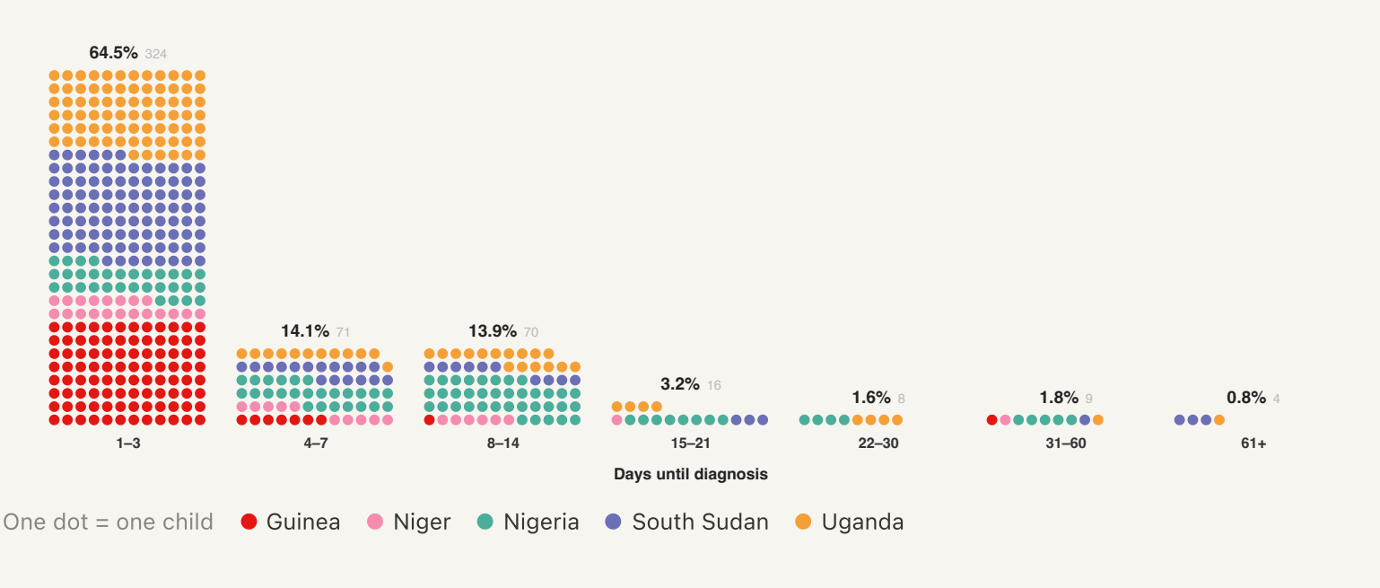
HIV-positive · **Under 2 years** · **Severely malnourished**

Both findings below held for each subgroup: diagnosis came within 3 days for the clear majority, and the scoring box was the dominant trigger.

FINDING 01 · SPEED

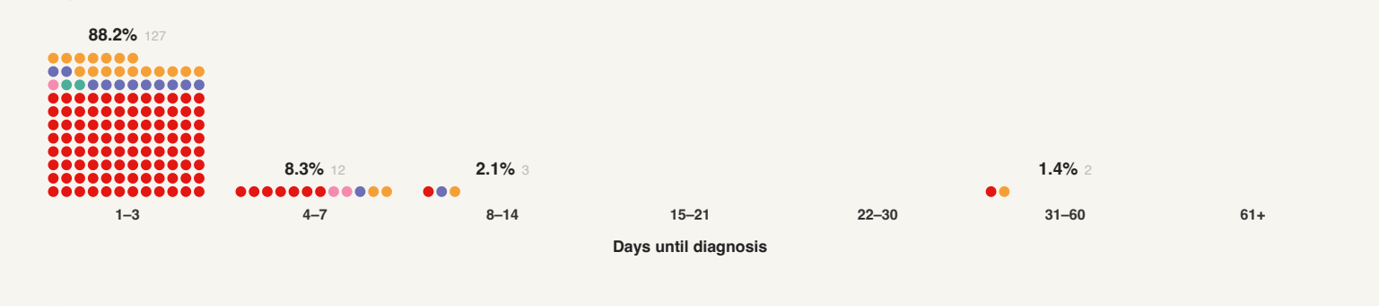
Most children diagnosed within 3 days of using the WHO algorithm

All children

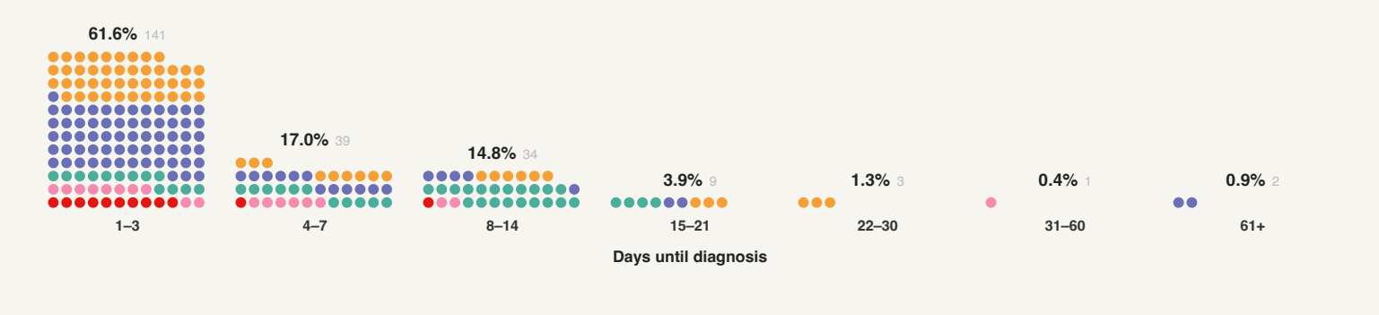


BY HIGH-RISK SUBGROUP

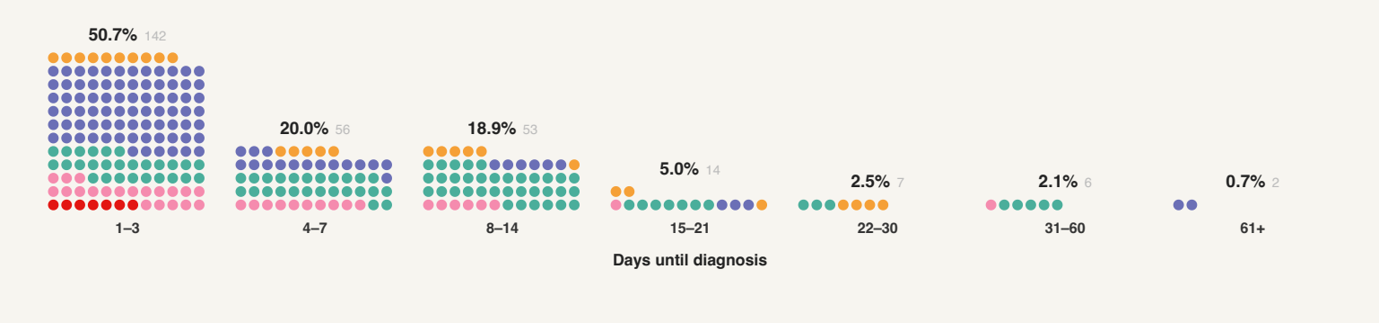
HIV-positive 144 children



Under 2 years 229 children



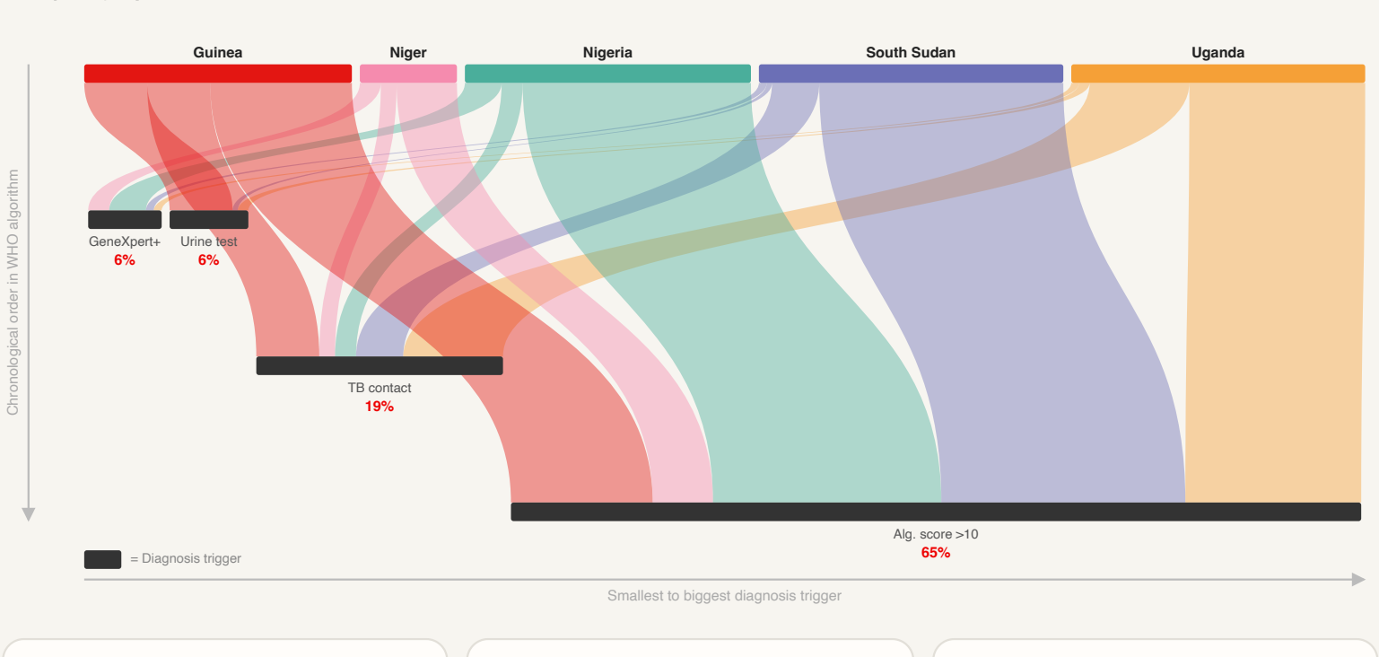
Severely malnourished 280 children



FINDING 02 · WHAT TRIGGERED DIAGNOSIS

Diagnosis was mostly triggered by the scoring box in the WHO algorithms

All children



65%
Algorithm score > 10

19%
Household TB contact

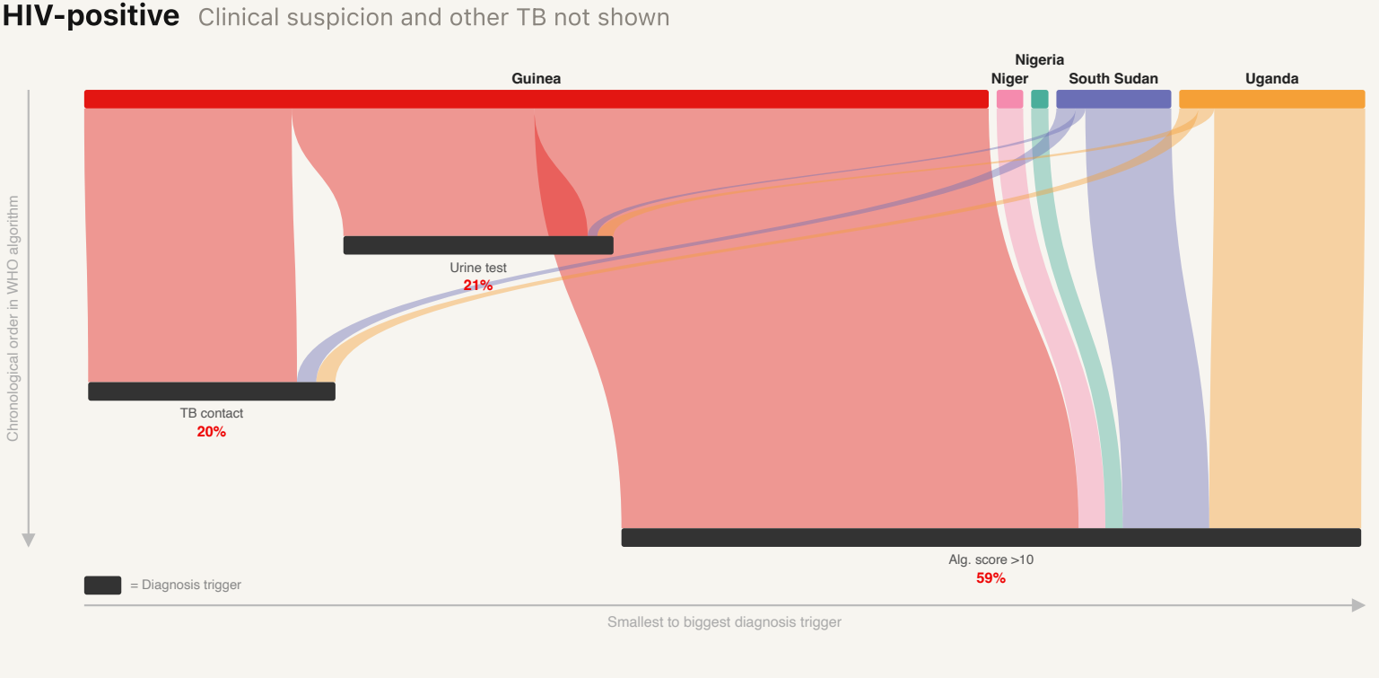
6%
GeneXpert positive

6%
Urine LF-LAM positive

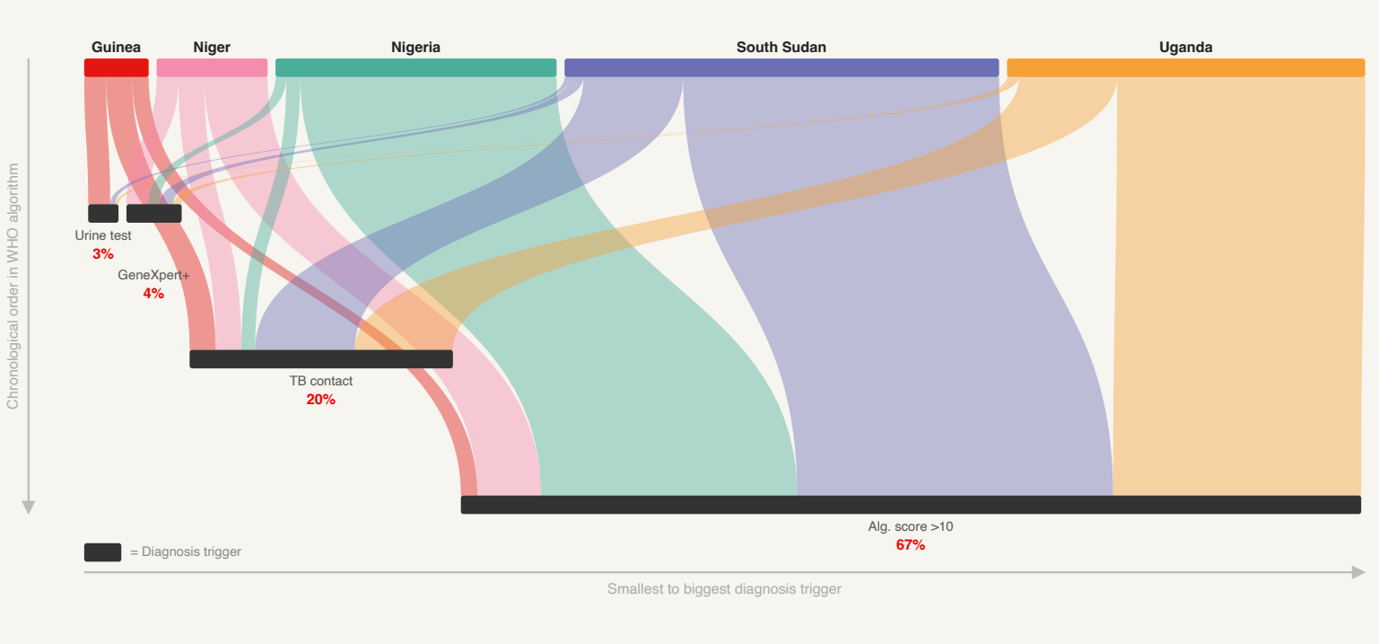
The remaining 4% came from clinical suspicion and other forms of TB. The scoring box triggered almost two thirds of diagnoses; laboratory tests together accounted for just **12%**.

BY HIGH-RISK SUBGROUP

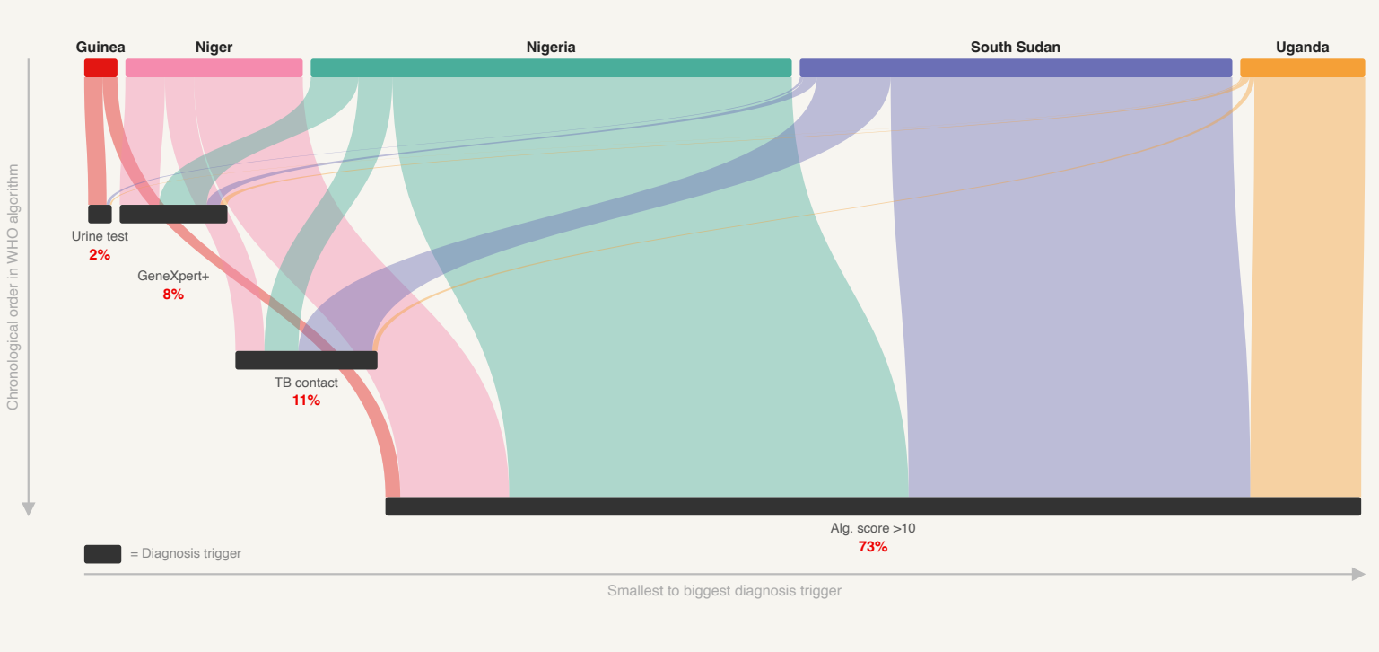
HIV-positive Clinical suspicion and other TB not shown



Under 2 years Clinical suspicion and other TB not shown



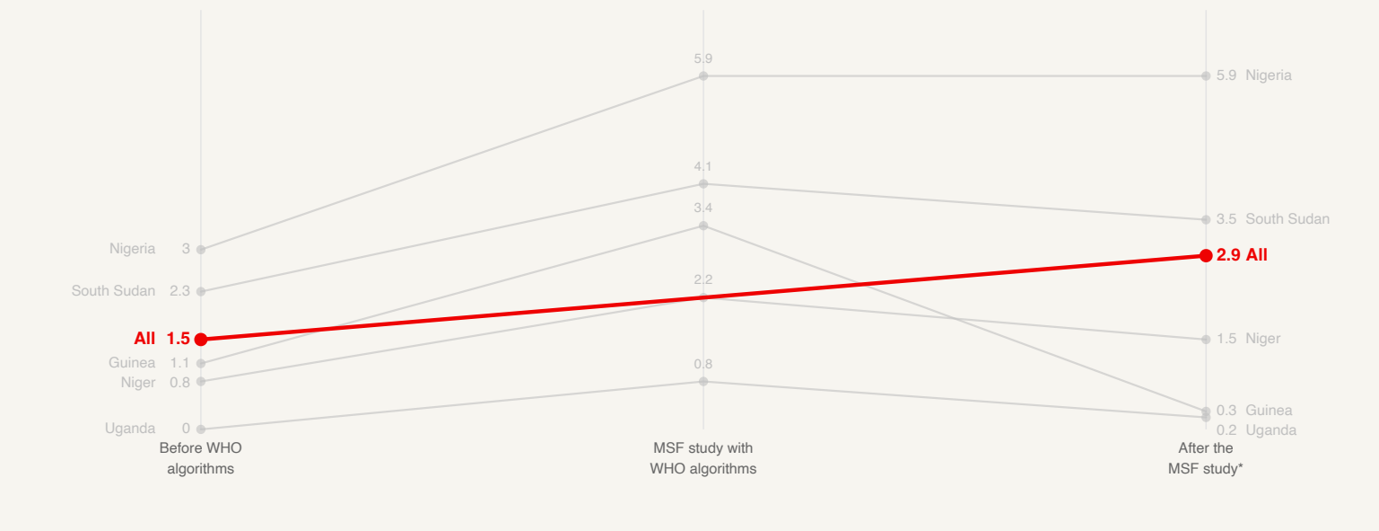
Severely malnourished Clinical suspicion and other TB not shown



FINDING 03 · DIAGNOSIS RATES

WHO algorithm use doubled TB diagnosis rates in children

Percent (%) of children diagnosed and admitted with TB, at three stages: before the WHO algorithms, inside the MSF study, and after the MSF study.



*Uganda never implemented the WHO algorithms into their national policies.

Across all five study countries, the proportion of children initiated on TB treatment doubled after the introduction of the algorithms, from 1.5% to 2.9%.

THE FEEDBACK

Clinicians say diagnosis with WHO algorithms is more *accessible and effective.*

DIAGNOSTIC SHIFT

"The WHO algorithms have streamlined the diagnostic and treatment decision-making approach for TB in children — a process that had long remained challenging. They allow us improve the prognosis for children with TB by facilitating **early diagnosis**."

Dr Moussa Mamane Oumarou Farouk
MSF TB Focal Point, Niger

"The algorithm has **opened our eyes to diagnose and treat TB in children** who would have been misdiagnosed."

Dr Chol
MSF Clinician, South Sudan

"Before, health workers relied on coughing... **The WHO algorithms now tell us not to rely on coughing — there are other signs for TB as well.**"

Dr Angeline Dore
MSF TACTiC Focal Point, Guinea

"Previously, there were patients who fell into a borderline category. We suspected TB, but the tests did not show positive results... **This algorithm has helped us a lot in such cases.**"

Dr Saima Umar
Head of Paediatrics, Gujranwala Teaching Hospital, Pakistan

IMPACT

"Thanks to these algorithms, we have **diagnosed tuberculosis earlier and saved more children** in health facilities with limited resources."

Dr Ibrahima Barry
Local Principal Investigator, TB ALGO PED study, Guinea

"Our findings prove that WHO's treatment decision algorithms, for which we do not need lab test results to initiate TB treatment in children, work in real-world settings and can potentially **save many more children's lives** if implemented."

Dr Helena Huerga
Principal Investigator, TB ALGO PED study, MSF

"The WHO's recommendations serve as a **crucial tool to help health workers diagnose TB in children in all settings**, even when there is limited access to laboratory tests and X-ray..."

Dr Gabriella Ferlazzo
TB Medical Advisor, MSF Access Campaign

"It has also **significantly reduced delays in starting treatment** in children. TB can spread quickly and lead to serious infections. Any delay in starting treatment can worsen their condition."

Dr Saima Umar
Head of Paediatrics, Gujranwala Teaching Hospital, Pakistan

"Using the algorithm is **easy, feasible and acceptable** to follow in the Malakal Project."

Dr Lazro Fidelle Nyikayo
MSF TB ALGO PED Study Coordinator, South Sudan

IMPLEMENTATION GAP

"**The science is clear** — what's now missing is **the political will to put it into practice.**"

Dr Helena Huerga
Principal Investigator, TB ALGO PED study, MSF

"In an already underfunded TB response, **children are pushed further to the back of the line** when services are disrupted by aid cuts, conflict, or displacement."

Dr Cathy Hewson
MSF TB Advisor

"For algorithms to be sustained in our hospitals, **regular monitoring and follow-up is important.**"

Dr Mohammed Bashir Abdullahi
MSF Medical Activity Manager, Nigeria

CALL TO ACTION

"We urge governments to adopt and put the WHO recommendations into practice... Governments must **leave no stone unturned to find every child with TB and treat them right away.**"

Dr Gabriella Ferlazzo
TB Medical Advisor, MSF Access Campaign

"The impact of implementing the WHO treatment decision algorithms in Niger has been **a catalyst for hope.**"

Dr Moussa Mamane Oumarou Farouk
MSF TB Focal Point, Niger

ADDRESSING CONCERNS

Frequently asked questions

An MSF nurse palpates 7-year-old Vedant's lymph nodes in Mumbai, India. His sister has drug-resistant TB. © Prem Hessenkamp.

How accurate are the WHO algorithms, and is there a risk of over-treatment?

Sensitivity · 90.5%	Specificity · 83.8%
Correctly identifies most children who *do* have TB — so fewer sick children are missed and go untreated.	Some children who *don't* have TB will still start TB treatment — a deliberate tradeoff. **With a disease this deadly, over-treating is less costly than missing a diagnosis.**

Aren't lab tests more sensitive than the WHO algorithms?

Not in young children. Current lab tests often fail to detect pediatric TB. WHO algorithms use clinical, radiographic, and contact-with-TB evidence.

Are health systems equipped to use WHO algorithms?

Yes. WHO algorithms are designed for all settings and reduce reliance on laboratory infrastructure and X-ray.

Is there enough evidence to support WHO algorithms?

Yes. Endorsed by the World Health Organization in 2022, WHO algorithms are backed by multi-country evidence showing strong diagnostic performance.

How complex is implementation?

No special devices or new equipment are needed. Rolling them out requires staff training, WHO algorithm integration into existing workflows and anticipating drug needs for treatment.

ACT NOW

The WHO algorithms save lives. *Children are waiting.*

Vaishnavi, 7, has DR-TB and interacts with MSF nurse Prachi in Mumbai, India. February 2022. © Prem Hessenkamp.

The WHO algorithms have been validated and can be adapted to local contexts today.

WHERE TO GO NEXT

WHO guidelines · tbksp.who.int/en/node/2032
About TACTiC · msf.org/tactic-tuberculosis-children
About TB ALGO PED · epicentre.msf.org/en/tb-algo-ped
Contact · tic-tb.children@paris.msf.org

READ THE INTERACTIVE VERSION

salkothari.github.io/msf-scrollytelling

Médecins Sans Frontières · TACTiC · Epicentre
Data: MSF TACTiC operational study (TB-ALGO-PED), Epicentre
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